HAna Exam Sheet 2025/26 - Back From The Dead

Reals, Sequences, and Series

Real Sequences

Definition: Bounds and Boundedness

Let $A \subseteq R$.
* $x \in \mathbb{R}$ is an **upper bound for** A if $x \geq a$ for all $a \in A$
* A is **bounded above** if there exists an upper bound $x \in \mathbb{R}$ for A.
* $s \in \mathbb{R}$ is a **least upper bound** or **supremum** for A if s is an upper bound for A, and $x \geq s$ whenever x is an upper bound for A.
* **Lower bounds** stuff is akin to this - if u don't know it go back to FPM
* A is **bounded** if it is bounded above and below

Axiom: The Completeness Axiom

Every bounded above non-empty $A \subseteq \mathbb{R}$ has a least upper bound in $\mathbb{R}$

Definition: Nested Sets

A sequence $(I_n)_{n\in\mathbb{N}}$ of sets is **nested** if $I_n \supseteq I_{n+1}$ for all $n \in \mathbb{N}$

Theorem: The Nested Interval Theorem

If $(I_n)_{n\in\mathbb{N}}$ is a nested sequence of nonempty closed bounded intervals, then the set $E := \bigcap_{n\in\mathbb{N}} I_n = \{x \in \mathbb{R} : x \in I_n \forall n \in \mathbb{N}\}$ is nonempty. Moreover, if $\lambda(I_n) \to 0$ as $n \to \infty$, where $\lambda(I_n)$ is the length of interval I_n, then E contains exactly one element.

Definition: Convergence

A real sequence $(a_n)_{n\in\mathbb{N}}$ **converges** to $a \in \mathbb{R}$ iff for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $|a_n - a| < \epsilon$ for all $n \geq N$. A sequence which does not converge to some real number **diverges**.
- We say $a_n \to \infty$ as $n \to \infty$ if for all $R \in \mathbb{R}$ there exists $N \in \mathbb{N}$ s.t. $a_n > R$ for all $n \geq \mathbb{N}$
- We say $a_n \to -\infty$ as $n \to \infty$ if for all $R \in \mathbb{R}$ there exists $N \in \mathbb{N}$ s.t. $a_n < R$ for all $n \geq \mathbb{N}$

Definition: Monotonicity

Let $(a_n)_{n\in\mathbb{N}}$ be a real sequence and $A := \{a_n : n \in \mathbb{N}\}$. Then:
(1) $(a_n)_{n\in\mathbb{N}}$ is **monotone non-decreasing** if $a_{n+1} \geq a_n$ for all $n \in \mathbb{N}$
(2) $(a_n)_{n\in\mathbb{N}}$ is **monotone non-increasing** if $a_{n+1} \leq a_n$ for all $n \in \mathbb{N}$
(3) $(a_n)_{n\in\mathbb{N}}$ is **bounded above/bounded below/bounded** if A is such
Monotone Convergence Theorem For Sequences: If $(a_n)_{n\in\mathbb{N}}$ is a real sequence which is monotone non-decreasing and bounded above; or monotone non-increasing and bounded below; then $(a_n)_{n\in\mathbb{N}}$ converges.

Major Definition: Cauchy On Sequences

A real sequence $(a_n)_{n\in\mathbb{N}}$ is **Cauchy** if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $|a_n - a_m| < \epsilon$ for all $n, m \geq N$. The *Cauchy Criterion* states that a real sequence is convergent if and only if it is Cauchy

Definition: Subsequences

Let $(a_n)_{n\in\mathbb{N}}$ be a real sequence. A **subsequence** of $(a_n)_{n\in\mathbb{N}}$ is a real sequence of the form $(a_{n_k})_{k\in\mathbb{N}}$ where for each $k \in \mathbb{N}$ there exists $n_k \in \mathbb{N}$ such that $n_1 < n_2 < \ldots < n_k < n_{k+1} < \ldots$. Ie, a subsequence is just some (possibly all) elements of a sequence taken in order.

Theorem: Bolzano-Weierstrass Theorem

Every bounded sequence of real numbers has a convergent subsequence

Series

Major Definition: Infinite Series

Let $(a_k)_{k\in N}$ be a real sequence. We form the **sequence of partial sums** $(s_n)_{n\in\mathbb{N}}$ by $s_n := \sum_{k=1}^{n} a_k$ for all $n \in \mathbb{N}$.
- We say the infinite series $\sum_{k=1}^{\infty} a_k$ **converges to** a if $\lim_{n\to\infty} s_n$ exists and equals a. Otherwise we say it **diverges**.

Definition: Absolute Convergence

Let $(a_n)_{n\in\mathbb{N}}$ be a real sequence.
(1) We say $\sum_{n=1}^{\infty} a_n$ **converges absolutely** if $\sum_{n=1}^{\infty} |a_n|$ converges
(2) We say $\sum_{n=1}^{\infty} a_n$ **converges conditionally** if it converges but $\sum_{n=1}^{\infty} |a_n|$ diverges.
- Absolute convergence implies convergence (the converse is not always true!)

Major Theorem: The Cauchy Criterion For Series

Let $(a_k)_{k=1}^{\infty}$ be a real sequence. Then $\sum_{k=1}^{\infty} a_k$ converges if and only if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $\left|\sum_{k=m+1}^{n} a_k\right| < \epsilon$ for all $n > m \geq N$.

Theorem: The Riemann Rearrangement Theorem

Let $(a_n)_{n\in\mathbb{N}}$ be a real sequence and suppose $\sum_{n=1}^{\infty} a_n$ is conditionally convergent. Then for *any* $s \in \mathbb{R}$, there exists a bijection $\sigma : \mathbb{N} \to \mathbb{N}$ such that the rearrangement $\sum_{n=1}^{\infty} a_{\sigma(n)}$ converges to s.

Note the RRT does not apply to absolutely convergent series. Indeed, for $\sum_{n=1}^{\infty} a_n$ absolutely convergent, and any bijection $\sigma : \mathbb{N} \to \mathbb{N}$, $\sum_{n=1}^{\infty} a_{\sigma(n)}$ is convergent and $\sum_{n=1}^{\infty} a_{\sigma(n)} = \sum_{n=1}^{\infty} a_n$

Subsets Of The Real Line

Definition: Open Sets

We say a set $U \subseteq \mathbb{R}$ is **open** if for all $x \in U$ there exists $r > 0$ such that $(x-r, x+r) \subseteq U$.
- The union of any collection of open sets is open
- The intersection of any finite collection of open sets is open
- Any nonempty open $U \subseteq \mathbb{R}$ is a finite or countable union of nonempty disjoint open intervals
- If $f : \mathbb{R} \to \mathbb{R}$ is continuous and $\lambda \in \mathbb{R}$, then $\{x \in \mathbb{R} : f(x) > \lambda\}$ is open.

Definition: Closed Sets

A set $F \subseteq \mathbb{R}$ is **closed** if its complement $\mathbb{R} \setminus F$ is open.
- The union of any finite collection of closed sets is closed
- The intersection of any collection of closed sets is closed
- If $f : \mathbb{R} \text{to} \mathbb{R}$ is continuous and $\lambda \in \mathbb{R}$, then $\{x \in \mathbb{R} : f(x) \leq \lambda\}$ is closed.

A set $F \subseteq \mathbb{R}$ is **sequentially closed** if for every convergent sequence $(x_n)_{n\in\mathbb{N}}$ s.t. $x_n \in F$ for all $n \in \mathbb{N}$, the limit $x := \lim_{n\to\infty} x_n$ satisfies $x \in F$.
- $F \subseteq \mathbb{R}$ is closed if and only if it is sequentially closed.

Definition: The Middle Thirds Cantor Set

The **Middle Thirds Canto Set** C is a closed set defined by the following recursive process:
(1) Divide $[0,1]$ into three equal parts and remove the open middle third interval $(\frac{1}{3}, \frac{2}{3})$.
(2) Divide the remaining two intervals into three equal parts and remove the open middle thirds intervals $(\frac{1}{9}, \frac{2}{9})$ and $(\frac{7}{9}, \frac{8}{9})$.
(3) Repeat this process ad infinitum.

At the nth stage of the construction we have 2^n closed intervals $(I_{n,k})_{k=1}^{2n}$ with $\lambda(I_{n,k}) = 3^{-n}$ for all $1 \leq k \leq 2^n$. C is then defined as $C = \bigcap_{n=1}^{\infty} C_n$ where $C_n := +\bigcup_{k=1}^{2^n} I_{n,k}$ for $n \in \mathbb{N}$. Each $I_{n,k}$ is a closed interval so their finite union C_n is closed, and C, the intersection of these closed C_n, is also closed.

Uniform Continuity, Uniform Convergence, Power Series

Uniform Continuity

Major Definition: Uniform Continuity

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. We say f is **uniformly continuous** if for every $\epsilon > 0$ there exists $\delta > 0$ such that $|f(x) - f(y)| < \epsilon$ for all $x, y \in I$ s.t. $|x - y| < \delta$. Note the key difference with regular continuity - here δ may only depend on ϵ only, not x.

Theorem: The Heine-Cantor Theorem

If $a < b$ and $f : [a,b] \to \mathbb{R}$ is continuous, then f is uniformly continuous.

Uniform Convergence

Definition: Pointwise Convergence

Let $E \subseteq \mathbb{R}$ be nonempty. A sequence of functions $f_n : E \to \mathbb{R}$ **converges pointwise on** E **to** $f : E \to \mathbb{R}$ if for all $x \in E$ and $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $|f_n(x) - f(x)| < \epsilon$ for all $n \geq N$.

Theorem: Pointwise Convergence Theorems

* The pointwise limit of bounded functions is not necessarily bounded
* The pointwise limit of continuous functions is not necessarily continuous. Ie, $\exists f_n, f : [0,1] \to \mathbb{R}$ s.t.
  - f_n is continuous for all $n \in \mathbb{N}$
  - $f_n \to f$ pointwise on $[0,1]$ as $n \to \infty$
  - f is not continuous
  - Example: $f_n(x) := 0$ if $0 \leq x \leq \frac{1}{2} - \frac{1}{2n}$; $nx - \frac{n-1}{2}$ if $\frac{1}{2} - \frac{1}{2n} \leq x \leq \frac{1}{2} + \frac{1}{2n}$; 1 if $\frac{1}{2} + \frac{1}{2n} \leq x \leq 1$
* The pointwise limit of differentiable functions is not necessarily differentiable. Ie, $\exists f_n, f : (-1,1) \to \mathbb{R}$ s.t.:
  - f_n is differentiable for all $N \in \mathbb{N}$
  - $f_n \to f$ pointwise on $(-1,1)$ as $n \to \infty$
  - f is not differentiable
  - Example: Let $f_n(x) := \sqrt{x^2 + \frac{1}{n}}$, limit is not diffable at 0.
* The pointwise limit of derivatives is not necessarily the derivative of the pointwise limit. Ie, $\exists f_n, f : (-1,1) \to \mathbb{R}$ s.t.
  - f_n is diffable for all $n \in \mathbb{N}$ and f is diffable
  - $f_n \to f$ pointwise on $(-1,1)$
  - $f'(0) \neq \lim_{n\to\infty} f_n'(0)$, ie $\left(\lim_{n\to\infty} f_n\right)'(0) \neq \lim_{n\to\infty} f_n'(0)$ - the derivate and ptwise lim operations do not commute.
  - Example: Let $f_n(x) = \frac{\sin(2\pi nx)}{2\pi n}, f_n' = \cos(2\pi nx), f = 0$

Definition: Graph and Vertical Neighbourhood

Let $f : E \to \mathbb{R}$. The **graph of** f is the set $\gamma(f) := \{(f, f(x) : x \in E)\} \subset \mathbb{R}^2$. For $\epsilon > 0$, the $e-$vertical neighbourhood of the graph of f is the set $\mathcal{N}(f, \epsilon) := \{(x,y) : x \in E, |y - f(x)| < \epsilon\} \subset \mathbb{R}^2$.

Major Definition: Uniform Convergence

Let $E \subseteq \mathbb{R}$ bd nonempty. A sequence of function s$f_n : E \to \mathbb{R}$ **converges uniformly on** E **to** $f :: E \to \mathbb{R}$ if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $|f_n(x) - f(x)| < \epsilon$ for all $x \in E, n \geq N$.
- Uniform convergence implies pointwise convergence (not the converse though! For counterexample use $f_n(x) := 1$ if $0 < x < \frac{1}{n}$; 0 if $\frac{1}{n} \leq x \leq 1$ and $f : (0,1] \to \mathbb{R}$ be $f(x) = 0$).

Theorem: Uniform Convergence Properties

Let $E \subseteq \mathbb{R}$ be nonempty, $f_n : E \to \mathbb{R}$ be a sequence of functions, and $f : E \to \mathbb{R}$.
- The following are equivalent:
  (1) $f_n \to f$ uniformly on E as $n \to \infty$
  (2) $\sup_{x\in E} |f_n(x) - f(x)| \to 0$ as $n \to \infty$
  (3) (Vertical neighbourhood test) For all $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $\gamma(f_n) \subset \mathcal{N}(f, \epsilon)$ for all $n \geq N$.
- (Sequence test for uniform convergence): Suppose $f_n \to f$ uniformly on E as $n \to \infty$. If $(x_n)_{n\in\mathbb{N}}$ is a real sequence with $x_n \in E$ for all $n \in \mathbb{N}$, then $|f_n(x_n) - f(x_n)| \to 0$ as $n \to \infty$.

Definition: Uniform Cauchy

Let $E \subseteq \mathbb{R}$ be nonempty and $f_n : E \to \mathbb{R}$ be a sequence of functions. We say $(f_n)_{n\in\mathbb{N}}$ is **uniformly Cauchy on** E if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ s.t. $|f_n(x) - f_m(x)| < \epsilon$ for all $x \in E$ and $m, n \geq N$.
* The Uniform Cauchy Criterion: $(f_n)_{n\in\mathbb{N}}$ converges uniformly on E iff $(f_n)_{n\in\mathbb{N}}$ is uniformly Cauchy on E.

Theorem: Uniform Limits Of Continuous And Differentiable Functions

- Let $I \subseteq \mathbb{R}$ be an interval and $f_n, f : I \to \mathbb{R}$ s.t. $f_n \to f$ uniformly on I as $n \to \infty$. For all $x_0 \in I$, if each f_n is continuous at x_0, then f is continuous at x_0.
- The uniform limit of diffable functions is not necessarily diffable, ie $\exists f_n, f : (-1,1) \to \mathbb{R}$ s.t.:
  - f_n is differentiable for all $n \in \mathbb{N}$
  - $f_n \to f$ uniformly on $(-1,1)$ as $n \to \infty$
  - f is not differentiable
  - Example: consider $f_n(x) := \sqrt{x^2 + \frac{1}{n}}$
- Let $I \subseteq \mathbb{R}$ be a bounded open interval and $f_n : I \to \mathbb{R}$ a sequence of diffable funcs s.t.:
  (1) There exists $x_0 \in I$ s.t. $(f_n(x_0))_{n\in\mathbb{N}}$ converges
  (2) There exists $g : I \to \mathbb{R}$ s.t. $f_n' \to g$ uniformly on I as $n \to \infty$.
  Then there exists diffable $f : I \to \mathbb{R}$ s.t. $f_n \to f$ uniformly on I as $n \to \infty$. Alternatively $(\lim_{n\to\infty} f_n)'(x) = \lim_{n\to\infty} f_n'(x)$ for all $x \in I$, ie differentiation commutes with taking the uniform limit under our conditions.

Definition: Uniform Convergence Of Series Of Functions